

YOKOTA AB, Japan

WMO: 476420

Lat: 35.749N

Lon: 139.349E

Elev: 472

StdP: 14.45

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 43313

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	27.5	6.4	7.8	40.5	9.6	9.1	38.8	24.7	44.6	22.3	43.8	3.6	280	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.2	(c) 93.2	(d) 78.6	(e) 91.1	(f) 77.8	(g) 88.1	(h) 76.4	(i) 79.9	(j) 89.4	(k) 78.9	(l) 87.4	(m) 77.9	(n) 85.5	(o) 9.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	144.6	83.6	76.7	141.5	83.2	75.4	135.4	82.6	44.0	90.3	42.7	87.9	41.6	85.5	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.6	(b) 18.7	(c) 16.1	(e) 21.2	(f) 97.8	(g) 2.0	(h) 2.6	(i) 19.8	(j) 99.7	(k) 18.6	(l) 101.3	(m) 17.5	(n) 102.7	(o) 16.1	(p) 104.6		
(4) 21.6	(5) 18.7	(6) 16.1	(7) 21.2	(8) 97.8	(9) 2.0	(10) 2.6	(11) 19.8	(12) 99.7	(13) 18.6	(14) 101.3	(15) 17.5	(16) 102.7	(17) 16.1	(18) 104.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	39.0	40.3	46.9	56.2	65.1	70.7	78.5	80.5	73.3	63.5	52.5	43.4		
	DBStd	15.14	4.13	4.88	5.89	6.20	4.89	4.93	5.38	4.68	5.61	5.19	5.20	4.71		
	HDD50	1014	342	275	136	17	0	0	0	0	0	0	33	211		
	HDD65	3534	806	691	561	270	58	9	0	0	5	89	375	670		
	CDD50	4398	1	4	40	202	469	621	882	946	700	420	107	6		
	CDD65	1446	0	0	0	5	62	180	418	482	255	44	0	0		
	CDH74	11645	0	0	2	56	372	957	3724	4714	1644	175	1	0		
	CDH80	4110	0	0	0	7	49	226	1454	1879	468	27	0	0		
	WSAvg	6.2	6.2	6.9	7.4	7.5	7.0	6.0	5.9	6.0	5.9	5.7	5.0	5.4		
	PrecAvg	58.9	2.0	1.9	3.3	4.1	5.3	6.7	6.7	7.0	8.9	7.6	3.2	2.2		
(15) Precipitation	PrecMax	82.2	4.9	3.7	6.3	10.2	10.5	15.5	12.8	15.8	15.9	24.4	11.5	6.3		
	PrecMin	41.2	0.2	0.1	1.1	1.2	2.2	2.8	2.7	1.7	1.5	0.9	0.1	0.4		
	PrecStd	8.2	1.4	1.1	1.3	2.2	2.1	2.6	3.1	4.0	4.3	5.3	2.2	1.3		
	PrecStd	8.2	1.4	1.1	1.3	2.2	2.1	2.6	3.1	4.0	4.3	5.3	2.2	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	64.7	71.9	79.2	84.3	89.9	96.3	96.8	91.7	83.7	71.6	64.1		
		MCWB	47.1	53.6	55.2	60.8	66.6	75.2	78.9	79.3	76.9	70.3	58.1	52.1		
	2%	DB	54.0	57.6	66.5	75.0	80.9	85.7	93.2	93.5	88.1	78.6	67.9	59.1		
		MCWB	43.1	45.8	52.9	58.8	64.8	72.2	78.2	79.1	75.6	67.1	56.9	48.2		
	5%	DB	51.5	53.7	62.5	71.5	78.4	82.3	91.0	91.4	85.7	75.2	64.7	55.8		
		MCWB	41.2	43.3	50.9	58.1	63.7	70.6	77.8	78.3	74.7	65.3	55.2	46.0		
	10%	DB	48.6	50.3	58.8	68.1	75.3	79.2	87.8	89.4	82.3	72.0	62.3	53.3		
		MCWB	38.9	40.9	48.9	56.2	62.8	69.2	76.7	77.5	72.6	63.4	54.0	44.9		
	0.4%	WB	50.1	55.8	58.6	66.6	71.0	77.8	81.3	81.8	79.0	72.6	62.8	55.0		
		MCDB	55.5	63.4	66.4	72.8	78.0	84.6	90.6	91.1	86.5	79.2	67.4	59.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.3	48.6	55.8	62.5	68.2	75.3	79.8	80.1	77.4	70.0	60.0	51.1		
		MCDB	50.9	56.0	63.3	70.5	76.0	82.7	89.3	89.8	83.9	75.9	64.3	55.6		
	5%	WB	42.5	44.9	52.9	60.2	66.5	73.0	78.6	79.2	76.2	67.7	57.6	48.2		
		MCDB	49.2	50.8	60.1	67.7	73.8	79.5	87.3	88.2	82.3	72.9	62.4	53.6		
	10%	WB	40.2	42.2	50.0	58.0	64.8	71.2	77.5	78.3	74.9	65.6	55.5	45.9		
		MCDB	46.9	48.3	57.0	65.5	71.8	76.6	85.4	86.3	80.4	70.4	60.2	51.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	19.0	19.1	19.3	17.4	13.7	13.8	14.2	13.3	13.8	16.7	18.8		
		MCDBR	22.9	24.0	24.7	24.1	22.3	18.6	18.3	17.6	17.1	17.6	20.6	21.9		
		MCWBR	14.5	15.1	14.7	12.2	10.6	8.4	7.3	6.8	7.5	9.2	12.1	13.5		
		MCDBR	20.5	21.0	22.3	20.2	18.3	15.0	16.6	16.1	15.1	14.6	16.7	18.6		
	5% WB	MCWBR	13.9	14.6	14.9	12.1	10.2	7.7	7.3	6.9	7.4	8.5	11.3	12.8		
		MCWBR	13.9	14.6	14.9	12.1	10.2	7.7	7.3	6.9	7.4	8.5	11.3	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.369	0.418	0.484	0.538	0.567	0.610	0.599	0.594	0.522	0.483	0.416	0.372		
	taud		2.252	2.114	1.957	1.870	1.855	1.818	1.895	1.896	2.052	2.084	2.196	2.279		
	Ebn at Noon		256	255	248	241	236	226	227	225	235	231	236	246		
	Edh at Noon		33	43	55	63	65	67	62	61	49	44	35	31		
(44) All-Sky Solar Radiation	RadAvg		882	1059	1315	1527	1638	1439	1530	1534	1209	978	812	771		
	RadStd		57	94	81	146	171	129	250	147	108	103	75	58		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page